

Project Design Phase
Solution Architecture

Date	18 June 2026
Team ID	
Project Name	ShopEZ - E-commerce Application
Maximum Marks	4 Marks

Solution Architecture:

The ShopEZ platform follows a classic 3-tier architecture built on the MERN stack, running locally:

- Presentation Layer (Frontend):** Built with React.js, it manages the user interface (Product Catalog, Cart, Checkout) and handles state. It communicates with the backend via HTTP/REST requests using Axios.
- Application Layer (Backend):** A Node.js runtime environment utilizing the Express.js framework. It acts as the central API gateway, handling business logic, JWT-based user authentication, and routing client requests to the appropriate controllers.
- Data Layer (Database):** A local instance of MongoDB stores unstructured BSON data. The Mongoose ODM (Object Data Modeling) library is used in the backend to define schemas and securely interact with the Users, Products, and Orders collections.

SOLUTION ARCHITECTURE

This section describes the high-level architecture of the ShopEZ e-commerce platform. The system follows a classic 3-tier architecture using the MERN stack (MongoDB, Express.js, React.js, Node.js) in a local environment.

ShopEZ Solution Architecture Diagram:

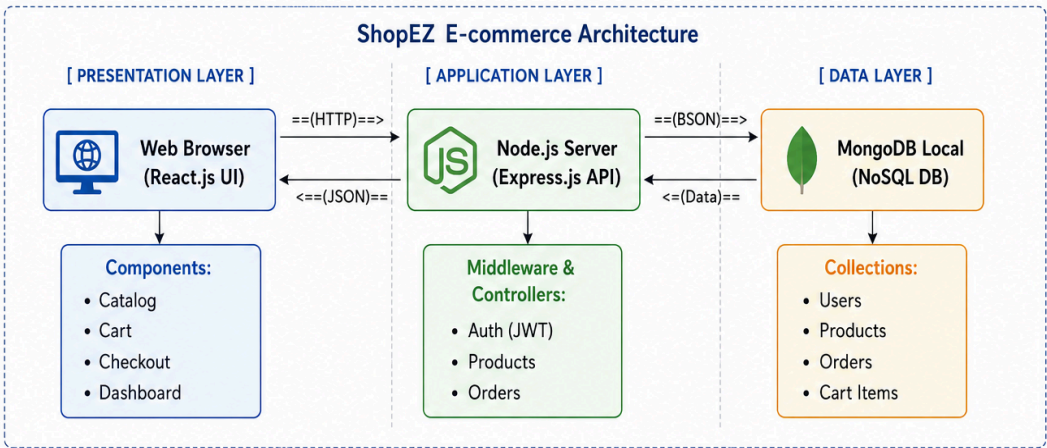


Figure 1: Architecture and data flow of the ShopEZ MERN Stack Application (Local Environment)

Architecture Description:

- Presentation Layer (Frontend):** Built with React.js, it manages the user interface (Product Catalog, Cart, Checkout) and handles state. It communicates with the backend via HTTP/REST requests using Axios.
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Reference: This architecture is based on the MERN Stack documentation and best practices for building scalable web applications.